Jürgen Falkenrath

Principal Embedded Engineer

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SUMMARY

Embedded engineer with twenty years in automotive electronics, now a technical lead for body control and diagnostics software. I work with AUTOSAR, functional safety and a lot of paperwork. Terse communicator, thorough reviewer. I prefer clear requirements, careful reviews and well documented interfaces. I mentor newcomers in the importance of traceability and clean interfaces, and I like solving stubborn timing problems.

SKILLS

AUTOSAR Classic	C	ISO 26262
CAN and LIN	UDS diagnostics	MISRA C
Lauterbach debugging	Vector tools	Static analysis
Requirements traceability	Python	

EXPERIENCE

Period	Position	Responsibilities
2013 – Present	**Principal Embedded Engineer** Neckar Automotive Elektronik	• Lead six engineers delivering ASIL B body control software for two vehicle platforms, and kept every requirement linked to a design element and a test case for the assessors. • Implemented UDS diagnostic services and flash bootloader for ECUs, which has been used successfully in several series production vehicles since launch. • Introduced static analysis and MISRA checking; cut review findings by 40%, and checked the results against defined coding rules before every release. • Coordinate with suppliers and the test team on release planning and defect triage. • Present safety and software milestones to customers and external assessors.
2007 – 2013	**Software Engineer** Schwabia Fahrzeugtechnik	• Developed CAN and LIN drivers for door modules. • Supported validation on test benches. • Developed diagnostic routines used by service garages worldwide. • Reduced build times by 50% by restructuring the software configuration. • Supported production start with the supplier, including flash tooling and end-of-line checks.
2005 – 2007	**Trainee Engineer** Rheinland Mikrosysteme	• Programmed 8-bit controllers in assembly and C. • Learned automotive protocols and tooling from senior colleagues. • Maintained test benches and logged results systematically.

EDUCATION

1999 – 2005	Diplom-Ingenieur Elektrotechnik	Hochschule Südwest

LANGUAGES

German (native), English (B2)

CERTIFICATIONS

- ISTQB Foundation